



YEAR 12 EXTENSION 2 MATHEMATICS JUNE 2006

42

NAME		RESULT	
DIRECTIONS	- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work. - Use black or blue pen only (<i>not pencils</i>) to write your solutions. - No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.		

Time allowed : 1 hour 10 minutes

Question 1

a) Find

(i) $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

2

(ii) $\int \frac{dx}{x^2 + 4x + 13}$

2

(iii) $\int \frac{1}{x^2 \sqrt{x^2 - 9}} dx$

4

b) Resolve $\frac{8}{(x+2)(x^2+4)}$ into partial fractions and hence find $\int \frac{8}{(x+2)(x^2+4)} dx$

4

c) Evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin \theta}$ using the substitution $t = \tan \frac{\theta}{2}$

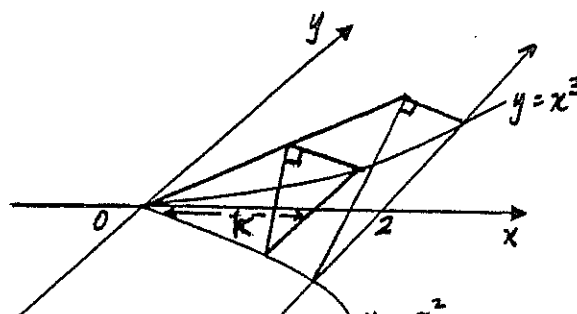
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Question 2 (Start a new page)

a) The base of a solid is the region in the xy plane enclosed by the curves $y = x^3$, $y = -x^2$ and the line $x = 2$. Each cross-section perpendicular to the x -axis is a right-angled isosceles triangle with the hypotenuse in the base of the solid.

(i) Show that the area of the triangular cross-section at $x = k$ is $\frac{1}{4}(k^4 + 2k^5 + k^6)$.

(ii) Hence find the volume of the solid.



2

2

Question 2 (continued)

- b) The area under the curve $y = \sin x$ between $x = 0$ and $x = \frac{\pi}{3}$ is rotated about the y axis to form a solid.

4

Use the method of cylindrical shells to find the volume of the resulting solid of revolution.

- c) Given $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$, use this result to find

4

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{e^x}{1+e^x} \tan^2 x dx$$

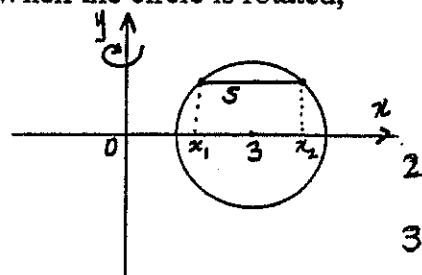
Question 3 (Start a new page)

- a) The circle $(x-3)^2 + y^2 = 4$ is rotated about the y -axis to form a ring. When the circle is rotated, the line segment S at height y sweeps out an annulus.

The x coordinates of the endpoints of S are x_1 and x_2 .

- (i) Show that the area of the annulus is equal to $12\pi\sqrt{4-y^2}$.

- (ii) Hence find the volume of the ring.



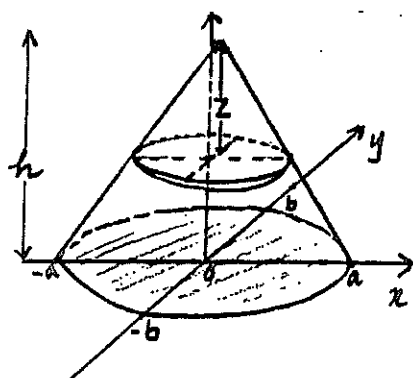
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- b) The solid is an elliptical cone of height h standing on a base with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Consider a slice of width δz parallel to the base at a distance z from the apex of the cone, and similar to the base.

You may assume that the area of an ellipse is given by πab where $2a$ and $2b$ are the lengths of the major and minor axes.

- (i) Show that the area of the cross-sectional area of the slice is $\frac{\pi ab z^2}{h^2}$.

- (ii) Find the volume of the elliptical cone.



3

2

- c) $I_n = \int_0^1 \frac{x^n}{1+x^2} dx$ where $n = 0, 1, 2, \dots$

- (i) Evaluate I_0

1

- (ii) Prove that $I_n + I_{n+2} = \frac{1}{n+1}$

2

- (iii) Show that $I_8 = \frac{1}{7} - \frac{1}{5} + \frac{1}{3} - 1 + \frac{\pi}{4}$

1

QUESTION 1 16 marks

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a) (i) $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

Let $u = \sqrt{x}$

$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$

$= \int 2 \sin u du$ ✓

$= -2 \cos u + C$ ✓

$= -2 \cos \sqrt{x} + C$

(-1) if not substituted back

(ii) $\int \frac{dx}{x^2 + 4x + 13}$

$= \int \frac{dx}{(x+2)^2 + 3^2}$ ✓

$= \frac{1}{3} \tan^{-1} \left(\frac{x+2}{3} \right) + C$ ✓

(iii) $\int \frac{1}{x^2 \sqrt{x^2 - 9}} dx$

$x = 3 \sec \theta$

$dx = 3 \sec \theta \tan \theta d\theta$ ✓

$= \int \frac{3 \sec \theta \tan \theta d\theta}{9 \sec^2 \theta \sqrt{9 \tan^2 \theta}}$ ✓

$= \frac{1}{9} \int \frac{1}{\sec \theta} d\theta$

$= \frac{1}{9} \int \cos \theta d\theta$

$= \frac{1}{9} \sin \theta + C$

$= \frac{1}{9} \sqrt{x^2 - 9}$ ✓

b) $\frac{8}{(x+2)(x^2+4)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+4}$

$8 = A(x^2+4) + (Bx+C)(x+2)$

$8 = (A+B)x^2 + (2B+C)x + 4A+2C$

Matching coeff of x^2 : $A+B=0$

$\therefore A = -B$

Coeff of x : $2B+C=0$... ①

Constants: $4A+2C=8$... ②

$\therefore B = -1, C = 2$ ✓

$A = 1$ ✓

$\therefore \frac{8}{(x+2)(x^2+4)} = \frac{1}{x+2} + \frac{2-x}{x^2+4}$

$T = \int \frac{1}{x+2} + \frac{2-x}{x^2+4} = \int \frac{1}{x+2} + \frac{2}{x^2+4} - \frac{x}{x^2+4}$

$= \log_e |x+2| + \tan^{-1} \frac{x}{2} - \frac{1}{2} \log |x^2+4|$ ✓

10T 14+14 - 42 TOTAL

c) $t = \tan \frac{\theta}{2}$

$\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2} = \frac{1}{2} (\tan^2 \frac{\theta}{2} + 1)$

$\therefore \frac{2dt}{t^2+1} = d\theta$ Limits $\theta = \frac{\pi}{3}, t = \frac{1}{\sqrt{3}}$
 $\theta = 0, t = 0$

$\int_0^{\frac{\pi}{3}} \frac{d\theta}{1+\sin \theta} = \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1+\frac{t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2}$ ✓

$= \int_0^{\frac{1}{\sqrt{3}}} \frac{1+t^2}{1+t^2+2t} \cdot \frac{2dt}{1+t^2}$

$= \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{(t+1)^2} dt$ ✓

$= \left[-\frac{1}{t+1} \right]_0^{\frac{1}{\sqrt{3}}}$ ✓

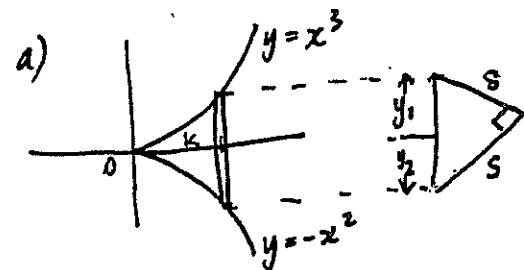
$= \frac{\sqrt{3}}{1+\sqrt{3}} = 1$

$= \frac{-1}{1+\sqrt{3}} \times \frac{1+\sqrt{3}}{1-\sqrt{3}} = \frac{-1+\sqrt{3}}{1-3}$ ①

$= \frac{-1+\sqrt{3}}{1-3}$

$= \frac{\sqrt{3}-1}{2}$

Question 2 (12 marks)



Hypotenuse length $= k^3 + k^2$ ✓

$2S^2 = (k^3 + k^2)^2$

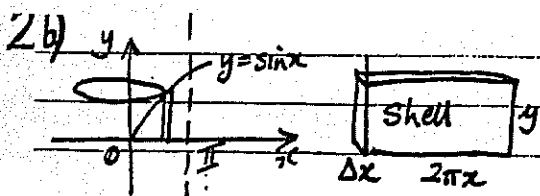
$= k^6 + 2k^5 + k^4$

Area of $\Delta = \frac{1}{2} S^2$ ✓

$= \frac{1}{4} (k^6 + 2k^5 + k^4)$

Volume $= \int_0^2 \frac{1}{4} (k^6 + 2k^5 + k^4) dk$

$= \frac{1}{4} \left[\frac{k^7}{7} + \frac{2k^6}{6} + \frac{k^5}{5} \right]_0^2$ ✓



$$A(x) = 2\pi xy$$

$$= 2\pi x \sin x$$

$$\text{Volume of Shell} = 2\pi x \sin x \Delta x$$

$$(ii) \text{ Total Volume} = 2\pi \int_0^{\pi/2} x \sin x dx$$

$$u = x \quad v' = \sin x$$

$$u' = 1 \quad v = -\cos x$$

$$V = 2\pi \left\{ [-x \cos x]_0^{\pi/2} + \int_0^{\pi/2} \cos x dx \right\}$$

$$= 2\pi \left\{ -\frac{\pi}{2} - 0 + [\sin x]_0^{\pi/2} \right\}$$

$$= \frac{\pi}{2} (3\sqrt{3} - \pi) \text{ or equivalent}$$



$$A(x) = \pi(x_2^2 - x_1^2)$$

$$= \pi(x_2 - x_1)(x_2 + x_1)$$

$$\text{Now } y^2 = (x-3)^2 = 4 - y^2$$

$$x-3 = \pm \sqrt{4-y^2}$$

$$x_2 = 3 + \sqrt{4-y^2}$$

$$x_1 = 3 - \sqrt{4-y^2}$$

$$\therefore A(x) = \pi(2\sqrt{4-y^2})(6)$$

$$= 12\pi \sqrt{4-y^2}$$

$$(ii) \text{ Total Vol} = \lim_{\Delta y \rightarrow 0} \sum_{-2}^2 12\pi \sqrt{4-y^2} \Delta y$$

$$V = 24\pi \int_0^2 \sqrt{4-y^2} dy$$

$$= 24\pi \int_0^{\pi/2} 2 \cos \theta \cdot 2 \cos \theta d\theta$$

$$= 48\pi \int_0^{\pi/2} \cos 2\theta + 1 d\theta$$

$$= 48\pi \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\pi/2}$$

$$= 48\pi \left(0 + \frac{\pi}{2} - 0 \right)$$

$$= 24\pi^2$$

$$\text{Alternatively } V = 12\pi \int_{-2}^2 \sqrt{4-y^2} dy$$

$$= 12\pi \times \frac{\pi 2^2}{2} \text{ Area of semicircle}$$

$$= 24\pi^2 \text{ of radius 2}$$

must state this

Note: $\int_0^2 \sqrt{4-y^2} dy$ is area of quadrant of circle, radius 2

$$(20) \int_{-\pi/4}^{\pi/4} \frac{e^x \tan^2 x}{1+e^x} dx$$

$$= \int_0^{\pi/4} \left[\frac{e^x}{1+e^x} + \frac{e^{-x}}{1+e^{-x}} \right] \tan^2(x) dx \checkmark$$

$$= \int_0^{\pi/4} \left[\frac{e^x}{1+e^x} + \frac{1}{e^x+1} \right] \tan^2 x dx \checkmark$$

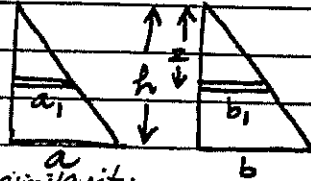
$$= \int_0^{\pi/4} \tan^2 x dx$$

$$= \int_0^{\pi/4} \sec^2 x - 1 dx \checkmark$$

$$= [\tan x - x]_0^{\pi/4} \checkmark$$

$$= 1 - \frac{\pi}{4}$$

3b)



Let a_1 and b_1 be semi axes of elliptical slice

By similarity,

$$\frac{z}{h} = \frac{a_1}{a}$$

$$\frac{b_1}{b} = \frac{z}{h}$$

$$\therefore a_1 = \frac{a \cdot z}{h}$$

$$b_1 = \frac{b \cdot z}{h}$$

$\therefore$ Cross sectional area of slice $= \pi a_1 b_1$

$$= \pi \frac{a \cdot z}{h} \cdot \frac{b \cdot z}{h} = \frac{\pi a b z^2}{h^2}$$

$$(ii) \text{ Vol of Slice} = \frac{\pi a b z^2}{h^2} \cdot \Delta z$$

$$\text{Total Vol} = \lim_{\Delta z \rightarrow 0} \sum_0^h \frac{\pi a b z^2}{h^2} \Delta z$$

$$= \frac{\pi a b}{h^2} \int_0^h z^2 dz$$

$$= \frac{\pi a b}{h^2} \left[\frac{z^3}{3} \right]_0^h$$

$$= \frac{\pi a b}{h^2} \cdot \frac{h^3}{3}$$

$$= \frac{\pi a b h}{3}$$

$$(i) I_0 = \int_0^1 \frac{1}{1+x^2} dx = [\tan^{-1} x]_0^1 = \frac{\pi}{4} \checkmark$$

$$(ii) I_n + I_{n+2} = \int_0^1 \frac{x^n}{1+x^2} + \frac{x^{n+2}}{1+x^2} dx$$

$$= \int_0^1 \frac{x^n(1+x^2)}{1+x^2} dx \checkmark$$

$$= \int_0^1 x^n dx$$

$$= \left[\frac{x^{n+1}}{n+1} \right]_0^1 = \frac{1}{n+1} \text{ as required} \checkmark$$

$$(iii) I_{n+2} = \frac{1}{n+1} - I_n \text{ Letting } n=6 \text{ we get}$$

$$\therefore I_7 = \frac{1}{7} - I_6 = \frac{1}{7} - \left(\frac{1}{5} - I_4 \right)$$

$$= \frac{1}{7} - \frac{1}{5} + \left(\frac{1}{3} - I_2 \right)$$

$$= \frac{1}{7} - \frac{1}{5} + \frac{1}{3} - \left(\frac{1}{1} - I_0 \right) = \frac{1}{7} - \frac{1}{5} + \frac{1}{3} - 1 + \frac{\pi}{4}$$